Aris Ozark, Dylan Thornburg

ECEN 133L-1

Thursday, April 3rd - 2:15 to 5:00

Lab 1 - Week 1

Lab #1: Impulse, Step, and Frequency Response

Objectives:

In this week's lab, our task is to run simple MATLAB scripts to create sampled signals, observe the effect of recursive filters, and become familiar with the impulse function. Throughout, we will increase our familiarity with course concepts and become more comfortable with using MATLAB for lab assignments.

Part 1: lab_1_s1.mlx

1. Read through the four sections of the script and determine what each instruction does.

Section 1: Creates and plots 3 vectors.
Section 2: Plots the vectors but isolates the specific points as well.
Section 3: Shows different multiplications of vectors.
- *p1 creates matrix of values where every possible combination of multiplying each element occurs*
- *p2 is a dot product operation*
- *p3 is element wise, each index multiplied by same index of other*
- *p4 squares the values in v2*

Section 4: Attempts to plot v1,v2 and v1,v3 but fails.

2. Run the first section and observe the results.

- **What values do you expect to see in the vectors v1, v2 and v3 based on the MATLAB statements?**
  - *V1 - [1,3,5,7,9]*
  - *V2 - [4,1,-2,-5,-8]*
  - *V3 - [-2.5, -2.0, -1.5, -1.0, -0.5, 0.0, 0.5]*

- **On the plot, what color is v1? What color is v2? What color is v3? What is represented on the horizontal scale?**
  - *V1 is blue, V2 is red, V3 is yellow.*

3. Modify the first section to use *stem* instead of *plot*.

a. Is it easier to see from this stem plot that you have functions of a discrete variable rather than functions of a continuous variable?

- *If you have only one vector to view, it is easy to observe, if there are more, it can become difficult to see them all.*

b. Is it easier to see the values of the elements of your three vectors?

- *Same case here. It is easy to see discrete values from the same vector in a stem plot, but viewing multiple can be difficult.*

c. How easy is it to see which discrete values belong to each vector?

- *It is not easy to see many values.*

4. Run the second section, which explicitly puts markers on the plots.

a. Is it easy to see from this plot that you have functions of a discrete variable rather than functions of a continuous variable?

- *It is easier to see discrete values since they are marked yet unobtrusive to the rest of the graph.*

b. How does the horizontal scale of this plot compare to the plot from the first section? What made the horizontal scale different?

- *The vector index starts at 0 on this scale vs the first. This is because we plotted 0: nvx (where x = 1, 2, or 3) - 1. To change, plot 1:nvx.*

5. Run the third section and note the results for each of the four multiplications.

a. What are the dimensions of each of the multiplication results? Which are matrices? Which are vectors? Which are scalars?

- *p1 yields a 5x5 matrix containing the values of every entry multiplied by every other entry. This is called an outer product operation.*
- *p2 yields a scalar value representing the dot product of v1 and v2*
- *p3 yields a five entry array containing an element-wise multiplication of v1 and v2.*
- *p4 yields a five entry array containing each element of v2 squared*

c. Use the three different types of multiplication that were used to multiply v1 and v2 to multiply vectors v2 and v3. Which ones cause a MATLAB error? Why? For the multiplication that does not produce an error, what are the dimensions of the resulting product?

- *The element-wise multiplication .* does not work since v2 and v3 are not the same size.*
- *The outer product operation yields a 5 by 7 matrix, corresponding the dimensions of v2 and v3 respectively*

6. Run the fourth section.

a. Why does one of the two plots generate an error message?

- *The vector sizes of V1 and V3 don't match.*

Part 2: lab_1_s2.mlx

1. Read through the three sections of the script and determine what each instruction does. Note that text following a "%" is a comment.

- *Section 1: Initializes our sampling frequency, time step, and signal frequency.*
- *Section 2: Initializes our n count and vector of angles A, then creates a digital output signal from these.*
- *Section 3: Does the same thing for section 2 but plays the sound the wave makes instead of plotting it.*

2. Run the complete script without any changes.

a. How many cycles of the cosine waveform are shown in the plot? (If you are not sure, note why you are not sure.)

- *Two cycles are shown.*

b. If you can identify one period of the waveform in the plot, then, based on the plot, what is the period in seconds for the waveform? How is this related to 1/f1?

- *2 milliseconds. 1/500 = 0.002 which is where we get this 2 ms number from.*

c. Based on your observations in (a) how many samples are in one period of the waveform? (Note that the plot has 32 samples.)

- *Sixteen samples are taken every period.*

d. How does the plot compare to what you would expect to see in a plot of a cosine function? Note any differences.

- *It's a digital recreation so it's being interpolated linearly, which makes it spiky.*

3. In the initialization section, change the frequency, f1, to 1500 and run the complete script again. Repeat steps 2a through 2e.

- *Cycles: Roughly 6. The graph looks extra weird and spiky so it's hard to see where the periods actually are.*
- *Period: About ⅔ ms*
- *Sample count per period: ~5.33*
- *This looks much less like a cosine function, probably because the sample rate isn't high enough for the frequency.*

a. How does the plot compare to the plot you saw in Step 2?

- *The plot looks extra weird and spiky so it's hard to see where the periods actually are.*

b. How does the sound compare to the sound you heard in Step 2?

- *It sounds higher pitched*

4. Repeat Step 3 for f1=3500.

- *Cycles: roughly 14 cycles are shown in the 4 ms graph shown.*

- *Period: T = 1/f, this comes out to 258 us.*
- *Samples: samples/cycle = 2.28*
- *Waveform: The wave appears jagged due to there not being many samples per signal. However, the playback is clear enough to play 3500 Hz.*
- *Comparisons: The sound is higher pitched and appears to be higher frequency than previous steps.*

5. Repeat Step 3 for f1=6500, f1=7500, and f1=8500. Notes any similarities to the results previously observed for the frequencies in Steps 2, 3, and 4.

F1 = 6500 Hz
Cycles: Roughly 6. The graph looks extra weird and spiky so it's hard to see where the periods actually are.
Period: About ⅔ ms
Sample count per period: ~5.33
This looks just like the plot of f1 = 1500.

F1= 7500 & F1 = 8500
Cycles: 2
Period: 2 ms
Sample count per period: 16
Both of these matched f1 = 500. The behavior seems mirrored about multiples of the sampling freq. (8000). We have unique plots from f1= 0-4000, but then from 4000-8000, it repearts the same values backwards. In short, this is aliasing, and the max frequency observed is found the when the sampling frequency is double the signal frequency.

Part 3: Impulse Filtering

3. Make a new section, section 3, to compare two similar filters and see how the response depends on one filter parameter. The filter parameters define the computation of the output as described in the background section of the lab.

- **Filter 1 is defined by *a1 = [1, -0.9];* and *b1 = 1*;**
- **Filter 2 is defined by *a2 = [1, -0.95];* and *b2 = 1;***
- **Describe the difference in the two output plots. Compare the index values at which each appears to reach zero based on the plot. Compare the minimum value of each.**

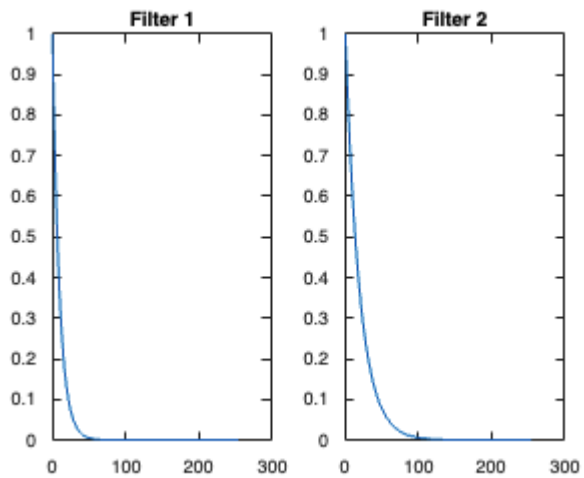

Figure 1: Plot of Filter 1 and Filter 2 in Section 3, impulse

- *Filter 1 exhibits a steep decline and appears to reach zero around index 50. The minimum value is 2.1470e-12, so very small!*
- *Filter 2 appears to have a much slower decay, and as a result, it has a much higher minimum value of 2.09e-6. Also, the plot appears to reach zero around index 100.*

4. Copy Section 3 to a new Section 4. Change filter 2 in Section 4 by setting *a2 = [1, 0.9];* and then run Section 4.

- **Describe the difference between the two output plots.**

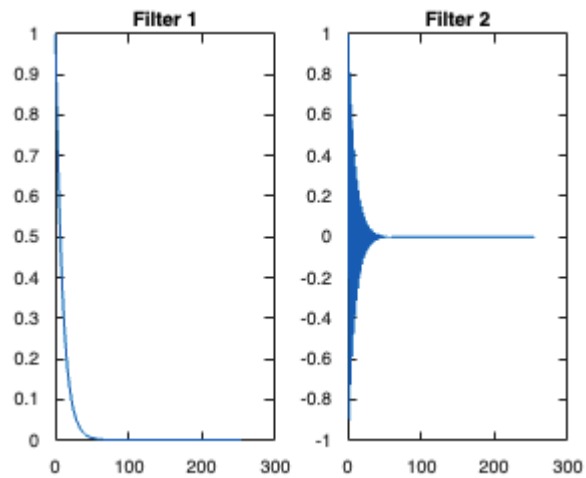

Figure 2: Plot of Filter 1 and Filter 2 in Section 4, impulse

- *Filter 2 exhibits a similar slope behavior to Filter 1, except the value switches sign every step. They appear to approach zero near the same index of ~50. The minimum values differ, with Filter 1 being at 2.15e-12 and Filter 2 being at -0.9. Also, the minimum absolute values of each filter are the same, near 2.15e-12.*

5. Copy Section 3 to a new Section 5. Change filter 2 in Section 5 by setting *a2 =b1;* and *b2 =a1;* and then run Section 5.

- **Describe the difference between the two output plots.**

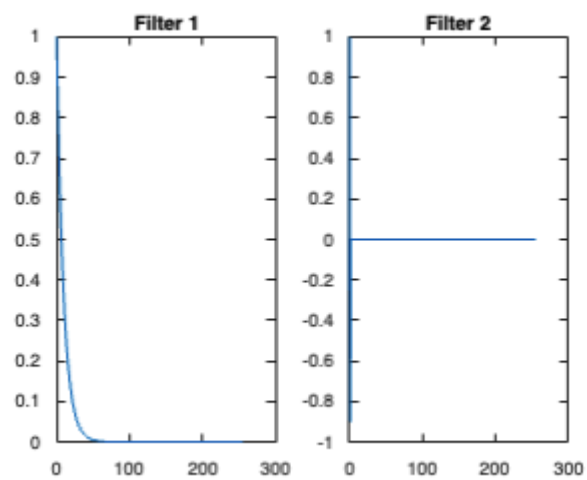

Figure 3: Plot of Filter 1 and Filter 2 in Section 5, impulse

- *Filter 2 rapidly oscillates between 1 and -0.9 before staying at zero after index 2. Filter 1 remains to approach zero after index 50. Similarly, to previous parts, the minimum values are as follows:*

• Compare the minimum value of each output.
Filter 1: 2.147e-12
Filter 2: -0.9
• Compare the minimum of the absolute value of each output.
Filter 1: 2.147e-12
Filter 2: 2.147e-12

Part 4: Square Wave Filtering

1. Modify Section 2 to make the input be one cycle of a square wave using
   *x = square((2*pi/256)*nv);*

4. In Section 3, compare the two outputs.

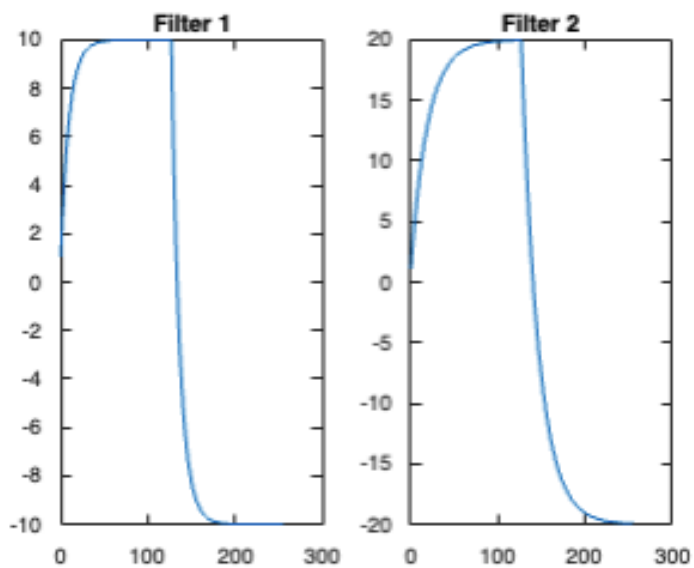

Figure 4: Plot of Filter 1 and Filter 2 in Section 3, square

- *Filter 1 arrives at its final value extremely quickly, only taking about 50 timesteps to arrive. Filter 2's magnitude is double that of filter 1, and it takes roughly double the amount of timesteps to arrive at its final value.*

- *Filter 2 has a larger maximum and smaller minimum because per the filter formula, we are adding 0.95*y(n-2) to the total output. Filter 1 has smaller value since the multiplier is smaller.*

• **Based on the plots, what are the maximum and minimum values of each output?**

Filter 1: max of 10, min of -10

Filter 2: max of 20, min of -20

• **Based on the plot, what is the approximate time index when the output value appears to reach its maximum value? How does this compare to the times in Part 3 for the impulse response to appear to have a value of zero?**

- *We observe the same behavior here. The filtered signal takes a similar amount of timesteps to reach its final value.*

5. In Section 4, compare the two outputs.

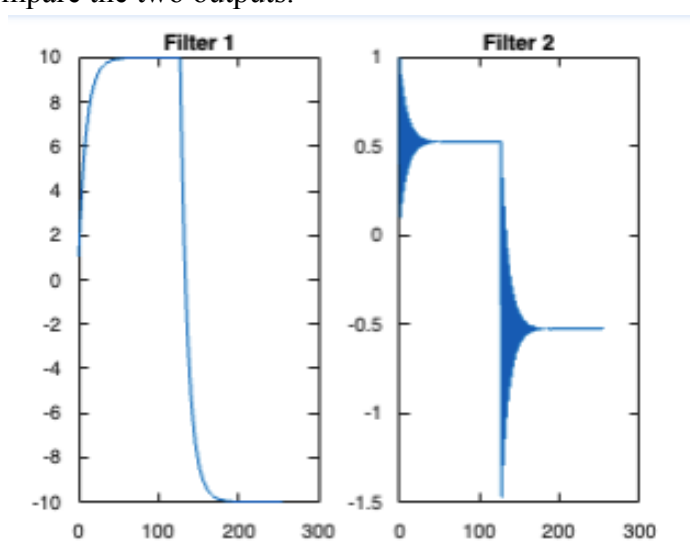

Figure 5: Plot of Filter 1 and Filter 2 in Section 4, square

• **Describe the difference between the two output plots.**

- *In Filter 2, the first oscillation occurs over a broader range (~2x) vs the other. They also settle on different values.*

• **Based on the plots, what are the constant values that each output attains after the response to an input value change settles to a steady value?**

- *Filter 1 exhibits the same behavior as in previous parts. Filter 2's first steady state is 0.5, and its final steady state is -0.5.*

6. In Section 5, compare the two outputs.

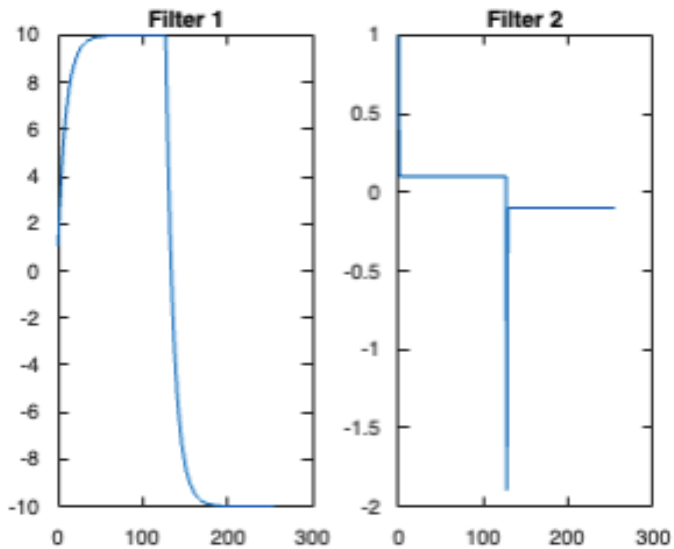

Figure 6: Plot of Filter 1 and Filter 2 in Section 5, square

- **Describe the difference between the two output plots.**
  - *Similar to Section 5 in Part 3, Filter 2 reaches its final value after only two timesteps in a dramatic jump. Filter 1 remains the same. Interestingly, Filter 2's magnitudes are quite small, only spanning 1 and -1.9 during both value changes.*
- **Based on the plots, what are the constant values that each output attains after the response to an input value change settles to a steady value?**
  - *Filter 1: First steady state is 10, second steady state is -10.*
  - *Filter 2: First steady state os 0.1, second is -0.1.*

Part 5: Sinusoidal Signal

1. **Modify Section 2 to make the input a sinusoidal signal using:**

   freq = 0.01;

   x = cos(2*pi*freq*nv);

4. **In Section 3, compare the two outputs.**

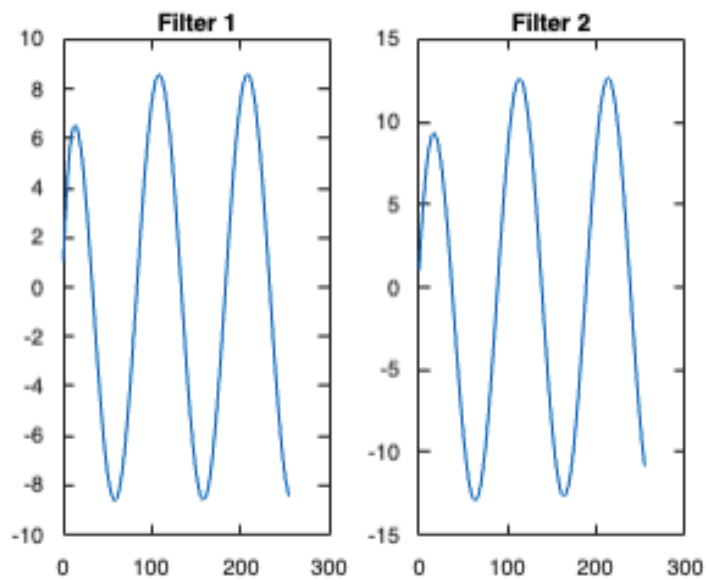

Figure 7: Plot of Filter 1 and Filter 2 in Section 3, sinusoid

- **Describe the difference between the two output plots. How do they look from time index 0 to time index of 100?**
  - *Filter 1 appears to be slightly delayed from filter 2, likely because it can not change as quickly*
- **After the time index of about 100, do they both appear to the be sinusoidal signals? Based on the plots, what is the amplitudes of each of each sinusoidal output after a time index of 100?**
  - *After index 100, they both appear to be regular sinusoidal signals. Filter 1 has an amplitude of 8.5 and Filter 2 has an amplitude of 12.6.*

5. In Section 4, compare the two outputs.

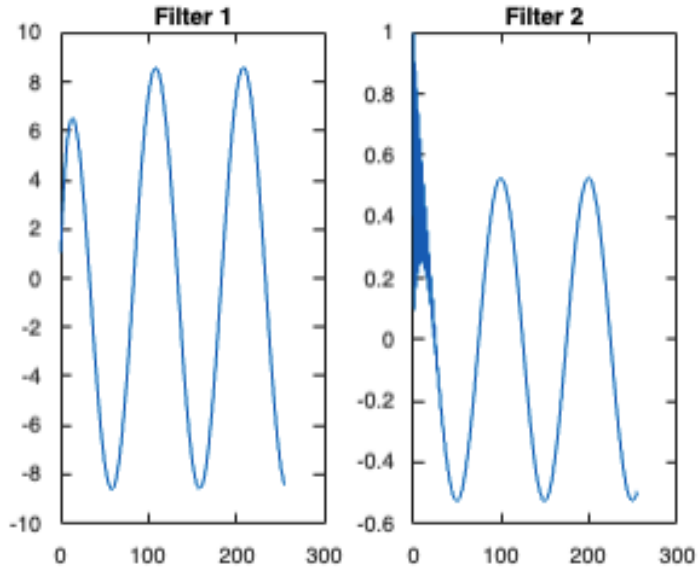

Figure 8: Plot of Filter 1 and Filter 2 in Section 4, sinusoid

- **Describe the difference between the two output plots. How do they look from time index 0 to time index of 100?**
  - *In the first 100 timesteps, Filter 1 appears approximately sinusoidal, but Filter 2 appears to be a rapidly oscillating, decaying waveform riding on top of a sinusoidal waveform.*
- **After the time index of about 100, do they both appear to the be sinusoidal signals? Based on the plots, what is the amplitudes of each of each sinusoidal output after a time index of 100?**

- *After 100 timesteps, both waveforms appear sinusoidal. The amplitude of FIlter 1 is 9, and that of Filter 2 is roughly 0.5.*

6. In Section 5, compare the two outputs.

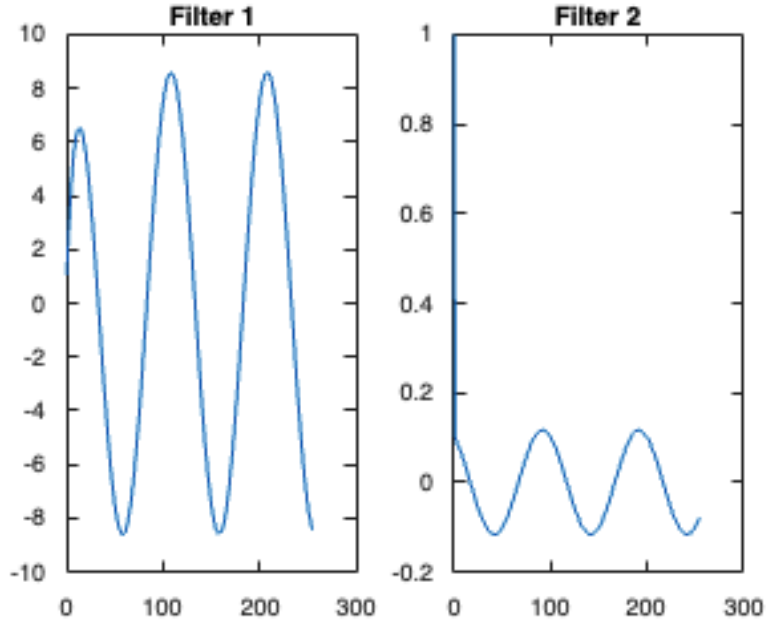

Figure 9: Plot of Filter 1 and Filter 2 in Section 5, sinusoid

- **Describe the difference between the two output plots. How do they look from time index 0 to time index of 100?**
  - *Filter one's wave grows into a sine wave (again), and filter two takes a sharp dive then becomes a low amplitude sine wave within the first 100 timesteps.*
- **After the time index of about 100, do they both appear to the be sinusoidal signals? Based on the plots, what is the amplitudes of each of each sinusoidal output after a time index of 100?**
  - *They do both appear sinusoidal. Filter 1's amplitude is again around 9, while filter 2 is roughly .1*

Conclusion:

In Part 1, we became familiar with different vector multiplication methods, their conditions for proper use, and how to troubleshoot their errors. In Part 2, we became familiar with the extent to which aliasing depends on sample rate. In particular, we learned that a sample rate of double the signal frequency provided the best assurance against aliasing. In Parts 3, 4, and 5, we learned how to plot filters using the MATLAB *filter* function. Different Filter designs can change how we see analog signals interpreted digitally. The behavior of transients, attenuation, and various other responses will assuredly play into future course material.